

Protocols for vehicle bus networks

Vehicle bus refers to the in-car network that connects the various electronic components inside a car including the power-train control module, body control modules, anti-lock braking system and so on. There are several technologies for this, the most popular being:

1. Controller area network (CAN): A multi-master asynchronous serial network protocol for high-reliability control applications
2. Local interconnect network (LIN): A low-speed, single-master, multiple-slave serial networking protocol
3. FlexRay: Next-generation, deterministic and fault-tolerant network protocol to enable high-bandwidth, safety-critical applications
4. Media-oriented system transport (MOST): A high-speed, synchronous multimedia network technology optimised by the automotive industry
5. Domestic digital bus (D2B): An isochronous high-speed, ring-based, fibre-optical communications technology
6. Radio frequency (RF): Radio frequency transmission, on/off keying or frequency-shift keying modulation
7. Ethernet: The ubiquitous networking protocol that is beginning to be used in automobiles as well

such as the UDF and KWP 2000 are also becoming popular.

Other common components in an automotive infotainment platform include hands-free telephony software frameworks, connectivity software frameworks, IEEE 1394 automotive software stacks, audio-video decoders and encoders, multimedia gateways, digital TV technologies like DVB-T/H, ISDB-T and ATSC/OpenCable, and AV/C General, FCP/CMP, IIDC controllers.

"Internet applications are set to make a major impact on car drivers and passengers during the next decade as we will undoubtedly witness new and innovative car-centric Internet applications and services. However, there will also be usage restrictions due to driver distraction issues while driving a car. Better interfaces with minimal driver distraction for Internet-based content will emerge in the next few years—some to be enabled by new infotainment systems and architectures. There is also a tremendous impetus coming from the mobile device world, with an explosion of applications related to mobility. Some of these applications will also find their way into cars," says Ganesh.

Power-train management

Power-train of an automobile is concerned with generation, transmission

and emission of energy, and includes the engine and transmission, wheels, suspension, drive shaft and exhaust system. It is typically managed using a power-train control module (PCM), which includes engine control and transmission control units. If engine is the heart of a car, PCM is the brain. PCM is perhaps one of the oldest applications of electronics in automobiles—although in its early years it was simply called an engine control unit.

"Automakers are under increasing pressure from both governments and consumers to improve overall fuel efficiency across their fleet (e.g., the US goal is 54.5 MPG by 2025) and lower polluting emissions. This is achieved by increasingly tight computerised control of the complete engine combustion process," says Ganesh.

A PCM often includes powerful processors (32-bit), sophisticated control strategies, smart sensors and actuators. It uses various sensors such as oxygen sensor, coolant sensor, mass air-flow sensor, air-intake sensor, crankshaft-angle sensor, throttle-position sensor, camshaft-angle sensor and knock sensor, to monitor the automobile's working and adjust the air/fuel mixture for maximum efficiency and least pollution.

Making a PCM requires cutting-edge design and development tools for application software develop-

ment, diagnostics, porting to different platforms, functional testing, re-engineering of legacy code, modelling of sensors, actuators and components, vehicle suspension controls and electronic power steering, calibration engineering, high-level synchronisation and so on. Tool chains like ETAS, INCA Instrumentation, CANalyzer, CANdela, CANoe, Pi auto simulator bench and static vehicle simulator, and open-/closed-loop LabCar are used. Protocols like CAN, LIN, J1850, OBD-II, EOBD, UDS and KWP2000 are popular.

Vehicle diagnostics

A lot of electronics goes into monitoring the health of vehicles. On-board electronics add self-diagnostic and reporting capabilities to modern automobiles. Earlier, diagnostic tools were limited to lighting up relevant indicators to warn the driver, but today's connected versions provide real-time information to drivers, fleet owners and rescue teams using standardised communication protocols and trouble codes, allowing rapid problem identification and rectification. Most modern tools can relay information to mobile devices.

A combination of on-board diagnostics, handheld scan tools, PC-based scan tools and analysis platforms, data loggers, etc is often used. Apart from the many analysers, scopes and meters installed in the vehicle to monitor engine, battery, air-conditioning, smoke and fluid levels, enthusiastic drivers often go in for additional instrumentation to monitor more parameters. These components are often managed by a diagnostic engine and advanced software controls.

Several standards and tool chains are available for design and development of diagnostic equipment, production testing, remote diagnostic interfaces and application-programming interfaces, etc. On-board diagnostics-II (OBD2) and Open diagnostics exchange (ODX) standards are popular today. These specify a data model to describe diagnostic data, including